Differentiation, variation and optimisation

Lecture and worked examples

Maths Explorer

Chapter path

Objectives. Calculate a derivative on its domain; study monotonicity and extrema; compare curves; optimise; use symmetries.

Prerequisites. Derivative at a point, algebraic calculation, sign of a quadratic, intervals.

Yearly reference point. 1-10 December: 8 hours, consolidation on the 15th and 17th

Derivative function and standard functions

A function is differentiable on an interval when it is differentiable at each of its points; its derivative associates with x the number $f'(x)$. The standard derivatives are: constant $\mapsto 0$; $x \mapsto 1$; $x^2 \mapsto 2x$; $x^3 \mapsto 3x^2$; $\frac{1}{x} \mapsto -\frac{1}{x^2}$ for $x \neq 0$; $\sqrt{x} \mapsto \frac{1}{2\sqrt{x}}$ for $x > 0$. For $n \in \mathbb{Z}$, $(x^n)' = nx^{n-1}$ on its domain of differentiability; for $n = 0$, the derivative is 0 directly.

Derivative function and standard functions

For $f(x) = x^{-3}$, the domain is $\mathbb{R} \setminus \{0\}$ and $f'(x) = -3x^{-4} = -\dfrac{3}{x^4}$. For $g(x) = \sqrt{x}$, the function exists at 0, its standard derivative only for $x > 0$.

Derivative function and standard functions

The domain of differentiability may be smaller than the domain of definition.

Derivative function and standard functions

Differentiate x^5, x^{-2} and $\sqrt{x}$, stating the domains.

Sum, product, reciprocal and quotient

For differentiable u and v, $(u+v)' = u' + v'$, $(\lambda u)' = \lambda u'$, $(uv)' = u'v + uv'$. Where $v \neq 0$, $(\frac{1}{v})' = -\frac{v'}{v^2}$ and $(\frac{u}{v})' = \frac{u'v - uv'}{v^2}$.

Sum, product, reciprocal and quotient

For $f(x) = \dfrac{x^2+1}{x-1}$, $x \neq 1$, $f'(x) = \dfrac{[2x(x-1)-(x^2+1)]}{(x-1)^2} = \dfrac{x^2-2x-1}{(x-1)^2}$. Keeping the brackets avoids changing the sign of a single term instead of the whole numerator.

Sum, product, reciprocal and quotient

The derivative of a product is not the product of the derivatives; the derivative of a quotient is not their quotient.

Sum, product, reciprocal and quotient

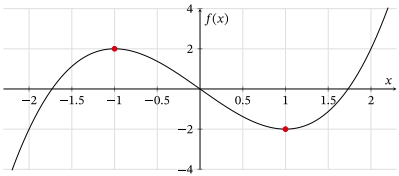
Differentiate $(x^2 + 1)(3x - 2)$, then $\dfrac{1}{x^2 + 1}$.

Sign of the derivative and extrema

On an interval I, if $f' \geq 0$, f is increasing; if $f' \leq 0$, it is decreasing; if $f' = 0$ everywhere, it is constant. A strictly positive derivative implies strict increase. If f has a local extremum at an interior point where it is differentiable, its derivative is zero there. The converse is false.

Sign of the derivative and extrema

$f(x) = x^3 - 3x$: $f'(x) = 3(x-1)(x+1)$. f increases on $(-\infty, -1]$, decreases on $[-1, 1]$, increases on $[1, +\infty)$; local maximum $f(-1) = 2$, local minimum $f(1) = -2$. Neither of these extrema is global on $\mathbb{R}$.



Sign of the derivative and extrema

The theorem applies on an interval. A function can decrease on two separate intervals without decreasing on their union.

Sign of the derivative and extrema

Study $f(x) = x^2 - 4x + 1$ on $[0, 3]$ and give its extrema on this interval.

Parity, inequalities and comparison

On a domain symmetric about 0, f is even if $f(-x) = f(x)$ and odd if $f(-x) = -f(x)$. The curve of an even function is symmetric about the y-axis; that of an odd function about the origin. To compare f and g, we study the sign of $f - g$, possibly with its derivative.

Parity, inequalities and comparison

For $x > 0$, $h(x) = x + \dfrac{1}{x} - 2$ has derivative $\dfrac{x^2 - 1}{x^2}$. It decreases up to 1 then increases; $h(1) = 0$, so $x + \dfrac{1}{x} \geq 2$. Equality is obtained only at 1.

Parity, inequalities and comparison

Check the domain before studying parity. The sign of $f' - g'$ tells us about the monotonicity of $f - g$, not directly about its sign.

Parity, inequalities and comparison

Study the parity of $x^4 + x^2$ and $x^3 - x$; prove $x^2 + 1 \geq 2x$.

Optimisation

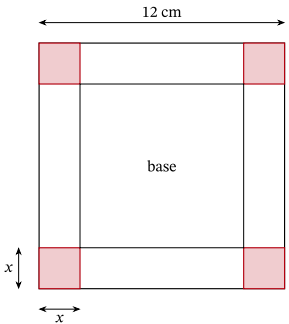
Optimising means defining the variable and its domain, expressing the quantity to optimise, then comparing all the candidate values, including the endpoints.

Optimisation

A box has a square base of side x and height $6 - x$ (the sum of the side and the height is 6), with $0 \leq x \leq 6$. Its volume $V(x) = x^2(6 - x)$ has derivative $V'(x) = 12x - 3x^2 = 3x(4 - x)$, positive on $[0, 4]$ and negative on $[4, 6]$. The maximum volume is $V(4) = 32$.

Optimisation

A numerical value produced by an algorithm is not a proof of a minimum or of convergence.

Optimisation

A rectangle of perimeter 12 m has width x.
Determine the maximum area by differentiation.